

SQL Analysis Findings

1. Overall Cancellation Performance

SQL analysis was performed on the combined 2009–2010 and 2010–2011 retail transaction data to measure overall cancellation activity.

Result

- Total transactions: **1,067,371**
- Total cancelled transactions: **19,494**
- Overall cancellation rate: **approximately 1.83%**

Business Interpretation

The analysis shows that cancellations represent a relatively small proportion of total transactions. However, the cancellation activity is large enough to warrant investigation because repeated cancellations can create operational effort, inventory disruption, and financial impact.

2. Year-wise Cancellation Performance

The cancellation performance was compared between the two available years.

Result

Year	Total Transactions	Cancelled Transactions	Cancellation Rate
2009–2010	525,461	10,206	~1.94%
2010–2011	541,910	9,288	~1.71%

Business Interpretation

Although transaction volume increased in 2010–2011, the cancellation rate decreased compared with 2009–2010. This indicates an improvement in the proportion of transactions being cancelled.

3. Monthly Cancellation Trend

Monthly analysis was performed to identify periods with higher cancellation activity.

Business Interpretation

The monthly analysis helps identify periods where cancellation activity increases and can support operational planning. These periods can be investigated further for possible relationships with product demand, inventory availability, or operational processes.

Important: The specific highest months should be taken directly from the SQL output rather than assumed.

4. Product Cancellation Analysis

Product-level analysis was performed to identify products associated with higher cancellation activity.

Business Interpretation

Products with consistently high cancellation counts can be prioritized for further investigation. Possible areas for investigation include inventory availability, product information accuracy, order processing, and demand fluctuations.

The SQL analysis provides the product-level evidence required to identify these products.

5. Country Cancellation Analysis

Cancellation activity was analyzed across countries to identify geographic patterns.

Business Interpretation

Countries with higher cancellation activity can be investigated to determine whether operational or market-specific factors are contributing to cancellations.

The analysis can help management prioritize geographic areas for further investigation.

6. Transaction Value Analysis

Transaction value associated with cancellations was analyzed using the TransactionValue / Transaction_value fields from the two source tables.

Business Interpretation

Cancellation transaction value provides an additional perspective beyond cancellation counts. A product or period with fewer cancellations can still be important if the associated transaction value is high.

7. Negative Quantity Analysis

Negative-quantity transactions were analyzed as part of the data-quality and transaction-behavior assessment.

Business Interpretation

Negative quantities can be associated with cancellation-related transaction activity in retail datasets. However, negative quantity should not automatically be treated as identical to a cancellation.

The SQL validation was therefore performed to compare negative quantities with the project's Cancellation_Flag.

8. Data Quality Findings

The analysis identified several data-quality considerations:

- Missing Customer IDs are present in the dataset.

- Missing product descriptions are present.
- At least one invalid InvoiceDate value (b) was identified during Excel analysis.
- Transaction-value fields required careful handling during SQL import and analysis.
- The dataset contains cancellation-related records with negative quantities.

Business Interpretation

These data-quality issues should be considered when designing reporting and operational systems. Missing or invalid data can reduce the reliability of customer-level, product-level, and time-based analysis.

9. Key Business Insights

Based on the SQL and Excel analysis:

1. Cancellation activity represents approximately **1.83%** of total transactions.
2. The cancellation rate decreased from approximately **1.94% in 2009–2010** to **1.71% in 2010–2011**.
3. Cancellation patterns vary across months, products, and countries.
4. Cancellation transaction value provides an additional measure of business impact beyond cancellation volume.
5. Data-quality issues such as missing Customer IDs, missing descriptions, and invalid dates need to be considered when interpreting results.

10. Business Recommendations

Based on the analysis, the retailer can consider:

1. Monitor cancellation KPIs

Create a regular dashboard tracking cancellation volume, cancellation rate, and cancellation value.

2. Investigate high-cancellation products

Review inventory, product information, and order-processing issues for products with consistently high cancellation activity.

3. Monitor high-risk periods

Use monthly cancellation trends to identify periods requiring additional operational attention.

4. Improve data quality

Introduce validation rules for customer information, product descriptions, dates, and transaction values.

5. Prioritize by business impact

Use both cancellation count and cancellation value when deciding which products or markets require attention.